

Supplementary Information for

Quantitative prediction of rate constants and its
application to organic emitters

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Supplementary Method 1

Method of calculating rate constants

In this study, each type of rate constant was calculated using the following equations.^{1,2}

1. $S_1 \rightarrow S_0$ fluorescence, $k_F(S_1 \rightarrow S_0)$

$$k_F(S_1 \rightarrow S_0) = \frac{\{E(S_1) - E(S_0)\}^3}{3\pi\epsilon_0\hbar^4c_3} \mu(S_1-S_0)^2 \times \frac{n_{\text{ref}}(n_{\text{ref}}^2 + 2)^2}{9} \quad (\text{A1})$$

$E(S_1)$: energy level of S_1

$E(S_0)$: energy level of S_0

$\mu(S_1-S_0)$: transition electric dipole moment between S_1 and S_0

n_{ref} : refractive index of the surrounding medium ($n_{\text{ref}} = 1.8$)

ϵ_0 : vacuum permittivity

$\hbar$: Dirac constant (Planck constant divided by 2π)

c : speed of light

2. $T_n \rightarrow S_0$ phosphorescence, $k_{\text{Phos}}(T_n \rightarrow S_0)$ ($n, m \geq 1$)

$$k_{\text{Phos}}(T_n \rightarrow S_0) = \frac{\{E(T_n) - E(S_0)\}^3}{3\pi\epsilon_0\hbar^4c_3} \mu(T_n-S_0)^2 \times \frac{n_{\text{ref}}(n_{\text{ref}}^2 + 2)^2}{9} \quad (\text{A2})$$

$$\mu(T_n-S_0) = \{\mu(T_n) - \mu(S_0)\} \frac{\text{SOC}(S_0-T_n)}{E(T_n) - E(S_0)}$$

$$+ \sum_m \mu(S_m-S_0) \frac{\text{SOC}(S_m-T_n)}{E(T_n) - E(S_m)} + \sum_m \mu(T_n-T_m) \frac{\text{SOC}(S_0-T_m)}{E(S_0) - E(T_m)} \quad (\text{A3})$$

$E(T_n)$: energy level of T_n

$E(T_m)$: energy level of T_m

$E(S_m)$: energy level of S_m

$\mu(T_n-S_0)$: transition electric dipole moment between T_n and S_0

$\mu(T_n)$: permanent electric dipole moment of T_n

$\mu(S_0)$: permanent electric dipole moment of S_0

$\mu(S_m-S_0)$: transition electric dipole moment of S_m

$\mu(T_n-T_m)$: transition electric dipole moment between T_n and T_m

$\text{SOC}(S_0-T_n)$: S_0-T_n SOC

$\text{SOC}(S_m-T_n)$: S_0-T_n SOC

$\text{SOC}(S_0-T_m)$: S_0-T_m SOC

3. Energy-downhill $S_1 \rightarrow T_n$ intersystem crossing, $k_{\text{ISC}}(S_1 \rightarrow T_n)$, and energy-uphill $T_n \rightarrow S_1$ reverse intersystem crossing, $k_{\text{RISC}}(T_n \rightarrow S_1)$, where $E(S_1) > E(T_n)$

$$k_{\text{ISC}}(S_1 \rightarrow T_n) = \frac{2\pi}{\hbar} |\text{SOC}(S_1-T_n)|^2 \times \text{LSF}_{\text{ISC}}(E(S_1) - E(T_n)) \quad (\text{A4})$$

$$\text{LSF}_{\text{ISC}}(E(S_1) - E(T_n)) = \frac{1}{\pi} \frac{\gamma}{\{E(S_1) - E(T_n)\}^2 + \gamma^2} \quad (\text{A5})$$

$$k_{\text{RISC}}(T_n \rightarrow S_1) = \frac{1}{3} \times k_{\text{ISC}}(S_1 \rightarrow T_n) \exp[-\beta\{E(S_1) - E(T_n)\}] \quad (\text{A6})$$

$|\text{SOC}(S_1-T_n)|$: norm of the S_1 - T_n SOC

γ : half width of half maximum of the line-shape function, LSF_{ISC}

β : inverse temperature ($\beta^{-1} = k_B T$)

k_B : Boltzmann constant

T : temperature

We used $\text{FWHM} = 1000 \text{ cm}^{-1}$ ($\gamma = 500 \text{ cm}^{-1}$) according to the previous report.¹

4. Energy-uphill $S_1 \rightarrow T_n$ intersystem crossing, $k_{\text{ISC}}(S_1 \rightarrow T_n)$, and energy-downhill $T_n \rightarrow S_1$ reverse intersystem crossing, $k_{\text{RISC}}(T_n \rightarrow S_1)$, where $E(S_1) < E(T_n)$

$$k_{\text{RISC}}(T_n \rightarrow S_1) = \frac{1}{3} \times \frac{2\pi}{\hbar} |\text{SOC}(S_1-T_n)|^2 \times \text{LSF}_{\text{ISC}}(E(T_n) - E(S_1)) \quad (\text{A7})$$

$$k_{\text{ISC}}(S_1 \rightarrow T_n) = 3k_{\text{RISC}}(T_n \rightarrow S_1) \exp[-\beta\{E(T_n) - E(S_1)\}] \quad (\text{A8})$$

5. $S_1 \rightarrow S_0$ nonradiative decay, $k_{\text{NR}}(S_1 \rightarrow S_0)$

$$k_{\text{NR}}(S_1 \rightarrow S_0) = \sum_{\alpha} k_{\text{NR},\alpha}(S_1 \rightarrow S_0) \quad (\text{A9})$$

$$k_{\text{NR},\alpha}(S_1 \rightarrow S_0) = \frac{R_{\alpha}(S_1-S_0)}{\hbar} \text{LSF}_{\text{NR},\alpha} \quad (\text{A10})$$

$$R_{\alpha}(S_1-S_0) = \frac{\hbar^2 V_{\alpha}(S_1-S_0)^2}{\{E(S_1) - E(S_0)\}^2} \quad (\text{A11})$$

$$\begin{aligned} \text{LSF}_{\text{NR},\alpha} &= 2\pi \sinh\left(\frac{1}{2}\beta\hbar\omega_{\alpha}\right) \sum_{v_{\alpha}} \exp\left\{-\left(\frac{1}{2} + v_{\alpha}\right)\beta\hbar\omega_{\alpha}\right\} \\ &\times \frac{1}{\pi S} \left\{ \frac{v_{\alpha}\gamma}{\{E(S_1) - E(S_0) + \hbar\omega_{\alpha}\}^2 + \gamma^2} + \frac{(v_{\alpha} + 1)\gamma}{\{E(S_1) - E(S_0) - \hbar\omega_{\alpha}\}^2 + \gamma^2} \right\} \end{aligned} \quad (\text{A12})$$

$V_{\alpha}(S_1-S_0)$: vibronic coupling between S_1 and S_0 for the α^{th} vibrational mode

v_{α} : vibrational quantum number for the α^{th} vibrational mode

ω_{α} : angular frequency for the α^{th} vibrational mode

S : empirical parameter ($= 2.1947 \times 10^5$)

6. Energy-downhill $T_m \rightarrow T_n$ internal conversion, $k_{\text{IC}}(T_m \rightarrow T_n)$, and energy-uphill $T_n \rightarrow T_m$ internal conversion, $k_{\text{IC}}(T_n \rightarrow T_m)$, where $E(T_m) > E(T_n)$

$$k_{\text{IC}}(\text{T}_m \rightarrow \text{T}_n) = \sum_{\alpha} k_{\text{IC},\alpha}(\text{T}_m \rightarrow \text{T}_n) \quad (\text{A13})$$

$$k_{\text{IC},\alpha}(\text{T}_m \rightarrow \text{T}_n) = \frac{R_{\alpha}(\text{T}_m - \text{T}_n)}{\hbar} \text{LSF}_{\text{IC},\alpha} \quad (\text{A14})$$

$$R_{\alpha}(\text{T}_m - \text{T}_n) = \frac{\hbar^2 V_{\alpha}(\text{T}_m - \text{T}_n)^2}{\{E(\text{T}_m) - E(\text{T}_n)\}^2} \quad (\text{A15})$$

$$\begin{aligned} \text{LSF}_{\text{IC},\alpha} &= 2\pi \sinh\left(\frac{1}{2}\beta\hbar\omega_{\alpha}\right) \sum_{v_{\alpha}} \exp\left\{-\left(\frac{1}{2} + v_{\alpha}\right)\beta\hbar\omega_{\alpha}\right\} \\ &\times \frac{1}{\pi S} \left\{ \frac{v_{\alpha}\gamma}{\{E(\text{T}_m) - E(\text{T}_n) + \hbar\omega_{\alpha}\}^2 + \gamma^2} + \frac{(v_{\alpha} + 1)\gamma}{\{E(\text{T}_m) - E(\text{T}_n) - \hbar\omega_{\alpha}\}^2 + \gamma^2} \right\} \end{aligned} \quad (\text{A16})$$

$$k_{\text{IC}}(\text{T}_n \rightarrow \text{T}_m) = k_{\text{IC}}(\text{T}_m \rightarrow \text{T}_n) \exp[-\beta\{E(\text{T}_m) - E(\text{T}_n)\}] \quad (\text{A17})$$

$V_{\alpha}(\text{T}_m - \text{T}_n)$: vibronic coupling between T_m and T_n

7. $\text{T}_1 \rightarrow \text{S}_0$ nonradiative decay, $k_{\text{NR}}(\text{T}_1 \rightarrow \text{S}_0)$

$$k_{\text{NR}}(\text{T}_1 \rightarrow \text{S}_0) = \frac{2\pi}{\hbar} |\text{SOC}(\text{T}_1 - \text{S}_0)|^2 \times \frac{1}{3S} \text{LSF}_{\text{ISC}}(E(\text{T}_1) - E(\text{S}_0)) \quad (\text{A18})$$

$|\text{SOC}(\text{T}_1 - \text{S}_0)|$: norm of the $\text{T}_1 - \text{S}_0$ SOC

Supplementary Method 2

Method of calculating spin-orbit coupling, vibronic coupling constant, transition dipole moment, and permanent dipole moment

Spin-orbit coupling (SOC), vibronic coupling constant, and T₁-T₂ transition dipole moment were calculated using the following formulae.

1. Spin-orbit coupling between S₁ and T_n, |SOC(S₁-T_n)|

$$|\text{SOC}(S_1-T_n)| = \sqrt{\sum_{M_S=0,\pm 1} \left| \langle \Phi_{S_1} | \mathcal{H}_{\text{SOC}} | \Phi_{T_n}^{M_S} \rangle \right|^2} \quad (\text{B1})$$

$$\langle \Phi_{S_1} | \mathcal{H}_{\text{SOC}} | \Phi_{T_n}^{M_S} \rangle = \frac{\text{FSC}^2}{2} \sum_i \sum_A Z_A^{\text{eff}} \langle \Phi_{S_1} | \frac{\mathbf{r}_i - \mathbf{R}_A}{|\mathbf{r}_i - \mathbf{R}_A|^3} \times \nabla | \Phi_{T_n}^{M_S} \rangle \cdot \mathbf{s}_i \quad (\text{B2})$$

Φ_{S_1} : electronic wave function of S₁ state

$\Phi_{T_n}^{M_S}$: electronic wave function of T₁ state

FSC: fine-structure constant

Z_A^{eff} : effective nuclear charge of the Ath nucleus

$\mathbf{r}_i$: cartesian coordinates of the i^{th} electron,

$\mathbf{R}_A$: cartesian coordinates of the Ath nucleus,

$\mathbf{s}_i$: spin operator of the electron i

Φ_{S_1} and $\Phi_{T_n}^{M_S}$ were calculated with Gaussian 16 Rev C01 program package.³ The integral $\langle \Phi_{S_1} | \frac{\mathbf{r}_i - \mathbf{R}_A}{|\mathbf{r}_i - \mathbf{R}_A|^3} \times \nabla | \Phi_{T_n}^{M_S} \rangle$ was calculated with the method proposed by McMurchie and Davidson.⁴

2. Vibronic coupling constant between S₁ and S₀ for the α^{th} vibrational mode, $V_\alpha(S_1-S_0)$

$$V_\alpha(S_1-S_0) = - \sum_i \sum_A Z_A \frac{\mathbf{e}_{\text{MW},A}^{(\alpha)}}{\sqrt{M_A}} \cdot \langle \Phi_{S_1} | \frac{\mathbf{r}_i - \mathbf{R}_A}{|\mathbf{r}_i - \mathbf{R}_A|^3} | \Phi_{S_0} \rangle \quad (\text{B3})$$

Z_A : nuclear charge of the Ath nucleus

M_A : mass of the Ath nucleus

$\mathbf{e}_{\text{MW},A}^{(\alpha)}$: the Ath element of the eigenvector of the α^{th} vibrational mode

Φ_{S_1} : electronic wave function of S₁

Φ_{S_0} : electronic wave function of S₀

$\mathbf{r}_i$: cartesian coordinates of the i^{th} electron,

$\mathbf{R}_A$: cartesian coordinates of the Ath nucleus

Φ_{S_1} and Φ_{S_0} were calculated with Gaussian 16 Rev C01 program package.³ The integral $\langle \Phi_{S_1} | \frac{\mathbf{r}_i - \mathbf{R}_A}{|\mathbf{r}_i - \mathbf{R}_A|^3} | \Phi_{S_0} \rangle$ was calculated with the method proposed by McMurchie and Davidson.⁴ The formula for T_m-T_n vibronic coupling constant is obtained by replacing S₁ and S₀ with T_m and T_n, respectively.

3. T₁-T₂ transition dipole moment, $\mu(T_1-T_2)$

$$\mu(T_1-T_2) = -\langle \Phi_{T_1} | \mathbf{x} | \Phi_{T_2} \rangle \quad (\text{B4})$$

Φ_{T_1} : electronic wave function of T₁

Φ_{T_2} : electronic wave function of T₂

$\mathbf{x}$: point in the three-dimensional space

Φ_{T_1} and Φ_{T_2} were calculated with Gaussian 16 Rev C01 program package.³ The integral $\langle \Phi_{T_1} | \mathbf{x} | \Phi_{T_2} \rangle$ was calculated with the method proposed by McMurchie and Davidson.⁴

4. S₁-S₀ transition dipole moment and permanent dipole moment of S₀, T₁, and T₂ were calculated with Gaussian 16 Rev C01 program package³.

Supplementary Method 3

Method of calculating excited-state populations and the number of emitted photons

Transient photoluminescence decay curve is numerically calculated by the following kinetic equations for S_0 , S_1 , T_1 , and T_2 populations. The kinetic equations were solved numerically using our own code.

$$\begin{aligned} \frac{d}{dt}[S_1](t) = & -\{k_{ISC}(S_1 \rightarrow T_1) + k_{ISC}(S_1 \rightarrow T_2) + k_F(S_1 \rightarrow S_0) + k_{NR}(S_1 \rightarrow S_0)\}[S_1](t) \\ & + k_{RISC}(T_1 \rightarrow S_1)[T_1](t) + k_{RISC}(T_2 \rightarrow S_1)[T_2](t) \end{aligned} \quad (C1)$$

$$\begin{aligned} \frac{d}{dt}[T_2](t) = & -\{k_{RISC}(T_2 \rightarrow S_1) + k_{IC}(T_2 \rightarrow T_1) + k_{Phos}(T_2 \rightarrow S_0) + k_{NR}(T_2 \rightarrow S_0)\}[T_2](t) \\ & + k_{ISC}(S_1 \rightarrow T_2)[S_1](t) + k_{IC}(T_1 \rightarrow T_2)[T_1](t) \end{aligned} \quad (C2)$$

$$\begin{aligned} \frac{d}{dt}[T_1](t) = & -\{k_{RISC}(T_1 \rightarrow S_1) + k_{IC}(T_1 \rightarrow T_2) + k_{Phos}(T_1 \rightarrow S_0) + k_{NR}(T_1 \rightarrow S_0)\}[T_1](t) \\ & + k_{ISC}(S_1 \rightarrow T_1)[S_1](t) + k_{IC}(T_2 \rightarrow T_1)[T_2](t) \end{aligned} \quad (C3)$$

$$\begin{aligned} \frac{d}{dt}[S_0](t) = & \{k_F(S_1 \rightarrow S_0) + k_{NR}(S_1 \rightarrow S_0)\}[S_1](t) \\ & + \{k_{Phos}(T_2 \rightarrow S_0) + k_{NR}(T_2 \rightarrow S_0)\}[T_2](t) \\ & + \{k_{Phos}(T_1 \rightarrow S_0) + k_{NR}(T_1 \rightarrow S_0)\}[T_1](t) \end{aligned} \quad (C4)$$

$$N(t) = \{k_F(S_1 \rightarrow S_0)[S_1](t) + k_{Phos}(T_1 \rightarrow S_0)[T_1](t) + k_{Phos}(T_2 \rightarrow S_0)[T_2](t)\} \quad (C5)$$

$$\Phi = \int N(t)dt \quad (C6)$$

$$\Phi_{Phos}(T_1) = \int k_{Phos}(T_1 \rightarrow S_0)[T_1](t)dt \quad (C7)$$

$$\Phi_{Phos}(T_2) = \int k_{Phos}(T_2 \rightarrow S_0)[T_2](t)dt \quad (C8)$$

$[S_0(t)]$: population of S_0 at time t

$[S_1(t)]$: population of S_1 at time t

$[T_1(t)]$: population of T_1 at time t

$[T_2(t)]$: population of T_2 at time t

$k_{ISC}(S_1 \rightarrow T_1)$: rate constant for the $S_1 \rightarrow T_1$ intersystem crossing

$k_{ISC}(S_1 \rightarrow T_2)$: rate constant for the $S_1 \rightarrow T_2$ intersystem crossing

$k_F(S_1 \rightarrow S_0)$: rate constant for the $S_1 \rightarrow S_0$ radiative decay

$k_{NR}(S_1 \rightarrow S_0)$: rate constant for the $S_1 \rightarrow S_0$ nonradiative decay

$k_{RISC}(T_1 \rightarrow S_1)$: rate constant for the $T_1 \rightarrow S_1$ intersystem crossing

$k_{RISC}(T_2 \rightarrow S_1)$: rate constant for the $T_2 \rightarrow S_1$ intersystem crossing

$k_{IC}(T_2 \rightarrow T_1)$: rate constant for the $T_2 \rightarrow T_1$ internal conversion

$k_{Phos}(T_2 \rightarrow S_0)$: rate constant for the $T_2 \rightarrow S_0$ radiative decay

$k_{NR}(T_2 \rightarrow S_0)$: rate constant for the $T_2 \rightarrow S_0$ nonradiative decay

$k_{IC}(T_1 \rightarrow T_2)$: rate constant for the $T_1 \rightarrow T_2$ internal conversion

$k_{Phos}(T_1 \rightarrow S_0)$: rate constant for the $T_1 \rightarrow S_0$ radiative decay

$k_{NR}(T_1 \rightarrow S_0)$: rate constant for the $T_1 \rightarrow S_0$ nonradiative decay

$N(t)$: the number of photons emitted at time t per unit time

Φ : the total PLQY

$\Phi_{\text{Phos}}(\text{T}_1)$: the contribution from $\text{T}_1 \rightarrow \text{S}_0$ radiative decay to Φ

$\Phi_{\text{Phos}}(\text{T}_2)$: the contribution from $\text{T}_2 \rightarrow \text{S}_0$ radiative decay to Φ

In the main text, we define the lifetime for the total radiative decay as $\tau_{\text{toR}} = 1/k_{\text{toR}}$ and mention that τ_{toR} is a simple extension of the averaged radiative decay time proposed by Yersin et al.³ Here, we discuss this point in detail. From Equation 9, τ_{toR} is written as

$$\tau_{\text{toR}} = \frac{\sum_{l \geq 1} [S_l] + \sum_{l \geq 1} [T_l]}{\sum_{n \geq 1} k_F(S_n \rightarrow S_0)[S_n] + \sum_{m \geq 1} k_{\text{Phos}}(\text{T}_m \rightarrow S_0)[T_m]} \quad (\text{C9})$$

The lifetimes for $S_n \rightarrow S_0$ fluorescence ($\tau_F(S_n \rightarrow S_0)$) and $\text{T}_m \rightarrow S_0$ phosphorescence ($\tau_{\text{Phos}}(\text{T}_m \rightarrow S_0)$) are written as $\tau_F(S_n \rightarrow S_0) = 1/k_F(S_n \rightarrow S_0)$ and $\tau_{\text{Phos}}(\text{T}_m \rightarrow S_0) = 1/k_{\text{Phos}}(\text{T}_m \rightarrow S_0)$, respectively ($n, m = 1, 2, 3, \dots$). Hence,

$$\tau_{\text{toR}} = \frac{\sum_{l \geq 1} [S_l] + \sum_{l \geq 1} [T_l]}{\sum_{n \geq 1} \frac{1}{\tau_F(S_n \rightarrow S_0)} [S_n] + \sum_{m \geq 1} \frac{1}{\tau_{\text{Phos}}(\text{T}_m \rightarrow S_0)} [T_m]} \quad (\text{C10})$$

When the excited singlet and triplet states are thermally equilibrated and $E(\text{T}_1) \leq E(\text{S}_1)$, $[S_n]$ and $[T_m]$ can be written in terms of $[T_1]$ and $S_n\text{-T}_1$ and $\text{T}_m\text{-T}_1$ energy differences

$$[S_n] = \frac{1}{3} [T_1] \exp\left(-\frac{E(S_n) - E(\text{T}_1)}{k_B T}\right) \quad (\text{C11})$$

$$[T_m] = [T_1] \exp\left(-\frac{E(\text{T}_m) - E(\text{T}_1)}{k_B T}\right) \quad (\text{C12})$$

Therefore,

$$\tau_{\text{toR}} = \frac{\sum_{l \geq 1} e^{-\frac{E(S_l) - E(\text{T}_1)}{k_B T}} + \sum_{l \geq 1} 3e^{-\frac{E(\text{T}_l) - E(\text{T}_1)}{k_B T}}}{\sum_{n \geq 1} \frac{1}{\tau_F(S_n \rightarrow S_0)} e^{-\frac{E(S_n) - E(\text{T}_1)}{k_B T}} + \sum_{m \geq 1} \frac{3}{\tau_{\text{Phos}}(\text{T}_m \rightarrow S_0)} e^{-\frac{E(\text{T}_m) - E(\text{T}_1)}{k_B T}}} \quad (\text{C13})$$

τ_{toR} is the population-weighted average of $\tau_F(S_n \rightarrow S_0)$ and $\tau_{\text{Phos}}(\text{T}_m \rightarrow S_0)$. When only S_1 and T_1 are considered, C13 can be written as

$$\tau_{\text{toR}} = \frac{e^{-\frac{E(\text{S}_1) - E(\text{T}_1)}{k_B T}} + 3}{\frac{1}{\tau_F(\text{S}_1 \rightarrow S_0)} e^{-\frac{E(\text{S}_1) - E(\text{T}_1)}{k_B T}} + \frac{3}{\tau_{\text{Phos}}(\text{T}_1 \rightarrow S_0)}} \quad (\text{C14})$$

Yersin et al. proposed the averaged radiative decay time (τ_{av}) under the assumption that S_1 and T_1 are thermally equilibrated

$$\tau_{\text{av}} = \frac{e^{-\frac{\Delta E(\text{S}_1 - \text{T}_1)}{k_B T}} + 1}{\frac{1}{\tau(\text{S}_1)} e^{-\frac{\Delta E(\text{S}_1 - \text{T}_1)}{k_B T}} + \frac{1}{\tau(\text{T}_1)}} \quad (\text{C15})$$

where $\tau(\text{S}_1)$ and $\tau(\text{T}_1)$ are the decay times for S_1 and T_1 , respectively, and $\Delta E(\text{S}_1 - \text{T}_1)$ is the $\text{S}_1\text{-T}_1$ energy gap. Note that the factor of 3 originating from the three triplet substates does not appear explicitly in the Yersin et al.'s formula (Equation C15). Comparing Equations C14 and

C15 shows that τ_{toR} defined in this study (Equation C14) is a simple extension of the Yersin et al.'s formula (Equation C15).

When $E(S_1) < E(T_1)$ (in the case of inverted singlet-triplet excited states), $[S_n]$ and $[T_m]$ can be written in terms of $[S_1]$ and S_n-S_1 and T_m-S_1 energy differences

$$[S_n] = [S_1] \exp\left(-\frac{E(S_n) - E(S_1)}{k_B T}\right) \quad (\text{C16})$$

$$[T_m] = 3[S_1] \exp\left(-\frac{E(T_m) - E(S_1)}{k_B T}\right) \quad (\text{C17})$$

Therefore,

$$\tau_{\text{toR}} = \frac{\sum_{l \geq 1} e^{-\frac{E(S_l) - E(S_1)}{k_B T}} + \sum_{l \geq 1} 3e^{-\frac{E(T_l) - E(S_1)}{k_B T}}}{\sum_{n \geq 1} \frac{1}{\tau_F(S_n \rightarrow S_0)} e^{-\frac{E(S_n) - E(S_1)}{k_B T}} + \sum_{m \geq 1} \frac{3}{\tau_{\text{Phos}}(T_m \rightarrow S_0)} e^{-\frac{E(T_m) - E(S_1)}{k_B T}}} \quad (\text{C18})$$

When only S_1 and T_1 are considered, C18 can be written as

$$\tau_{\text{toR}} = \frac{1 + 3e^{-\frac{E(T_1) - E(S_1)}{k_B T}}}{\frac{1}{\tau_F(S_1 \rightarrow S_0)} + \frac{3}{\tau_{\text{Phos}}(T_1 \rightarrow S_0)} e^{-\frac{E(T_1) - E(S_1)}{k_B T}}} \quad (\text{C19})$$

(C19) and (C14) are essentially the same.

In the main text, we define the total ISC k_{toISC} as

$$k_{\text{toISC}} = \sum_{n \geq 1} \left(\sum_{m \geq 1} k_{\text{ISC}}(S_n \rightarrow T_m) \right) \frac{[S_n]}{\sum_{l \geq 1} [S_l]} \quad (\text{C20})$$

When the excited singlet states are thermally equilibrated, from (C16) and (C20)

$$k_{\text{toISC}} = \sum_{n \geq 1} \left(\sum_{m \geq 1} k_{\text{ISC}}(S_n \rightarrow T_m) \right) \frac{e^{-\frac{E(S_n) - E(S_1)}{k_B T}}}{\sum_{l \geq 1} e^{-\frac{E(S_l) - E(S_1)}{k_B T}}} \quad (\text{C21})$$

Thus, (C20) contains the effect of thermal activation and deactivation between the excited singlet states on k_{toISC} . Likewise, when the triplet states are thermally equilibrated,

$$k_{\text{toRISC}} = \sum_{m \geq 1} \left(\sum_{n \geq 1} k_{\text{RISC}}(T_m \rightarrow S_n) \right) \frac{[T_m]}{\sum_{l \geq 1} [T_l]} \quad (\text{C22})$$

can be written as

$$k_{\text{toRISC}} = \sum_{m \geq 1} \left(\sum_{n \geq 1} k_{\text{RISC}}(T_m \rightarrow S_n) \right) \frac{e^{-\frac{E(T_m) - E(T_1)}{k_B T}}}{\sum_{l \geq 1} e^{-\frac{E(T_l) - E(T_1)}{k_B T}}} \quad (\text{C23})$$

(C22) contains the effect of thermal activation and deactivation between the triplet states on k_{toRISC} .

Supplementary Table 1 | Nuclear coordinates of S₁ geometry of BNOO in gas phase optimised at the TD-TPSSH/6-31G(d) level of theory.

Atom	Element symbol	x (Å)	y (Å)	z (Å)
1	C	-0.030591	4.836110	0.422831
2	C	0.357717	6.065813	0.949661
3	C	1.290846	6.110435	1.985771
4	C	1.859043	4.922663	2.462596
5	C	1.469632	3.692895	1.937428
6	C	-1.469632	-3.692895	1.937428
7	C	-1.859043	-4.922663	2.462596
8	C	-1.290846	-6.110435	1.985771
9	C	-0.357717	-6.065813	0.949661
10	C	0.030591	-4.836110	0.422831
11	C	0.477826	3.623503	0.940309
12	C	-0.477826	-3.623503	0.940309
13	C	-0.889482	1.530142	-3.127720
14	C	-1.347847	2.766185	-3.586806
15	C	-1.363796	3.878397	-2.741981
16	C	-0.912728	3.712275	-1.429831
17	C	0.912728	-3.712275	-1.429831
18	C	1.363796	-3.878397	-2.741981
19	C	1.347847	-2.766185	-3.586806
20	C	0.889482	-1.530142	-3.127720
21	N	0.000000	-2.419601	0.394075
22	C	0.458801	-2.480292	-0.959730
23	C	0.426667	-1.331804	-1.803832
24	B	0.000000	0.000000	-1.167356
25	C	-0.426667	1.331804	-1.803832
26	C	-0.458801	2.480292	-0.959730
27	N	0.000000	2.419601	0.394075
28	C	0.000000	0.000000	3.214189
29	C	0.066567	-1.210015	2.522821
30	C	-0.003217	-1.204099	1.109402
31	C	0.000000	0.000000	0.378468
32	C	0.003217	1.204099	1.109402
33	C	-0.066567	1.210015	2.522821
34	O	-0.928886	4.837814	-0.611243
35	O	0.928886	-4.837814	-0.611243
36	H	-0.075250	6.967106	0.528318
37	H	1.593150	7.068212	2.397928
38	H	2.620928	4.953113	3.235614
39	H	1.933347	2.776733	2.286134
40	H	-1.933347	-2.776733	2.286134
41	H	-2.620928	-4.953113	3.235614
42	H	-1.593150	-7.068212	2.397928
43	H	0.075250	-6.967106	0.528318
44	H	-0.913067	0.677366	-3.799834
45	H	-1.696034	2.869136	-4.611150
46	H	-1.701320	4.856271	-3.066955

47	H	1.701320	-4.856271	-3.066955
48	H	1.696034	-2.869136	-4.611150
49	H	0.913067	-0.677366	-3.799834
50	H	0.000000	0.000000	4.299906
51	H	0.179196	-2.141367	3.066313
52	H	-0.179196	2.141367	3.066313

Supplementary Table 2 | Nuclear coordinates of S₁ geometry of BNSS in gas phase optimised at the TD-TPSSH/6-31G(d) level of theory.

Atom	Element symbol	x (Å)	y (Å)	z (Å)
1	C	-4.859001	-0.658323	0.010570
2	C	-6.005399	-1.130224	0.659873
3	C	-5.895403	-1.861881	1.842254
4	C	-4.633090	-2.080683	2.404577
5	C	-3.489453	-1.590501	1.778118
6	C	3.398366	-1.802436	-1.663430
7	C	4.507981	-2.363325	-2.275492
8	C	5.803840	-2.072505	-1.815485
9	C	5.974023	-1.193705	-0.752583
10	C	4.861001	-0.617440	-0.122061
11	C	-3.583093	-0.907156	0.551974
12	C	3.544378	-0.934614	-0.554248
13	C	-1.462727	3.097165	-0.990891
14	C	-2.651878	3.553801	-1.546917
15	C	-3.747434	2.693649	-1.678241
16	C	-3.626169	1.365493	-1.237089
17	C	3.667300	1.430923	1.154184
18	C	3.760018	2.751101	1.641889
19	C	2.642551	3.571274	1.558308
20	C	1.451475	3.083892	1.002957
21	N	2.428237	-0.388755	0.081826
22	C	2.476997	0.947423	0.599294
23	C	1.317321	1.774226	0.504049
24	B	-0.002314	1.140859	-0.015799
25	C	-1.302197	1.767952	-0.509910
26	C	-2.447485	0.914751	-0.634691
27	N	-2.423094	-0.420037	-0.108955
28	C	0.001383	-3.217863	0.200497
29	C	1.217762	-2.515777	0.241788
30	C	1.195619	-1.122551	0.096895
31	C	-0.004397	-0.395924	0.019933
32	C	-1.208549	-1.122182	0.011671
33	C	-1.203543	-2.538501	0.063014
34	S	-4.978848	0.236814	-1.512802
35	S	5.116382	0.424878	1.265246
36	H	-6.979645	-0.918807	0.228830
37	H	-6.789127	-2.236524	2.332639

38	H	-4.537737	-2.614668	3.345750
39	H	-2.513936	-1.732474	2.231893
40	H	2.401530	-2.005684	-2.037414
41	H	4.368765	-3.015735	-3.132168
42	H	6.670337	-2.510137	-2.301129
43	H	6.969014	-0.935695	-0.402386
44	H	-0.607952	3.766884	-0.949849
45	H	-2.731999	4.580432	-1.895798
46	H	-4.680475	3.030639	-2.118164
47	H	4.699755	3.110360	2.049138
48	H	2.695771	4.594343	1.920204
49	H	0.586091	3.738744	0.962949
50	H	0.005789	-4.301725	0.263180
51	H	2.153995	-3.048076	0.375171
52	H	-2.136325	-3.088063	-0.007886

Supplementary Table 3 | Nuclear coordinates of S_1 geometry of BNSeSe in gas phase optimised using the TD-TPSSH method. For the H, B, C, and N atoms, the 6-31G(d) basis set was used. For the Se atoms, the Stuttgart/Dresden pseudopotentials and basis set (SDD) were used.

Atom	Element symbol	x (Å)	y (Å)	z (Å)
1	C	0.405028	4.837447	0.756781
2	C	1.171868	5.879949	1.285342
3	C	2.338372	5.604678	1.997899
4	C	2.760589	4.277484	2.150037
5	C	2.012980	3.237532	1.611024
6	C	-2.012980	-3.237532	1.611024
7	C	-2.760589	-4.277484	2.150037
8	C	-2.338372	-5.604678	1.997899
9	C	-1.171868	-5.879949	1.285342
10	C	-0.405028	-4.837447	0.756781
11	C	0.803417	3.496113	0.928709
12	C	-0.803417	-3.496113	0.928709
13	C	-0.900012	1.514751	-3.096470
14	C	-1.403539	2.728242	-3.562394
15	C	-1.484990	3.830002	-2.709591
16	C	-1.036708	3.691136	-1.385359
17	C	1.036708	-3.691136	-1.385359
18	C	1.484990	-3.830002	-2.709591
19	C	1.403539	-2.728242	-3.562394
20	C	0.900012	-1.514751	-3.096470
21	N	-0.043534	-2.423295	0.409126
22	C	0.489382	-2.493276	-0.925481
23	C	0.427074	-1.338011	-1.773343
24	B	0.000000	0.000000	-1.148036
25	C	-0.427074	1.338011	-1.773343
26	C	-0.489382	2.493276	-0.925481

27	N	0.043534	2.423295	0.409126
28	C	0.000000	0.000000	3.225231
29	C	0.000000	-1.212354	2.532475
30	C	-0.035167	-1.200687	1.117554
31	C	0.000000	0.000000	0.392758
32	C	0.035167	1.200687	1.117554
33	C	0.000000	1.212354	2.532475
34	Se	-1.242651	5.206223	-0.180429
35	Se	1.242651	-5.206223	-0.180429
36	H	0.851538	6.906424	1.132851
37	H	2.926326	6.419773	2.409263
38	H	3.688408	4.054438	2.668532
39	H	2.358246	2.212327	1.694538
40	H	-2.358246	-2.212327	1.694538
41	H	-3.688408	-4.054438	2.668532
42	H	-2.926326	-6.419773	2.409263
43	H	-0.851538	-6.906424	1.132851
44	H	-0.892706	0.657411	-3.763249
45	H	-1.745384	2.818045	-4.590412
46	H	-1.877652	4.781246	-3.053938
47	H	1.877652	-4.781246	-3.053938
48	H	1.745384	-2.818045	-4.590412
49	H	0.892706	-0.657411	-3.763249
50	H	0.000000	0.000000	4.310809
51	H	0.036833	-2.153693	3.070726
52	H	-0.036833	2.153693	3.070726

Supplementary Table 4 | Nuclear coordinates of S₁ geometry of BNTeTe in gas phase optimised using the TD-TPSSH method. For the H, B, C, and N atoms, the 6-31G(d) basis set was used. For the Te atoms, the Stuttgart/Dresden pseudopotentials and basis set (SDD) were used.

Atom	Element symbol	x (Å)	y (Å)	z (Å)
1	C	-4.776822	0.826524	-0.835490
2	C	-5.696138	1.352599	-1.752086
3	C	-5.258996	1.958018	-2.929218
4	C	-3.888258	2.009073	-3.214218
5	C	-2.968384	1.474101	-2.321646
6	C	2.968384	1.474101	2.321646
7	C	3.888258	2.009073	3.214218
8	C	5.258996	1.958018	2.929218
9	C	5.696138	1.352599	1.752086
10	C	4.776822	0.826524	0.835490
11	C	-3.392862	0.893659	-1.104589
12	C	3.392862	0.893659	1.104589
13	C	-1.562406	-3.112009	0.812702
14	C	-2.798431	-3.576708	1.257576
15	C	-3.899881	-2.720693	1.288627

16	C	-3.745937	-1.394460	0.841903
17	C	3.745937	-1.394460	-0.841903
18	C	3.899881	-2.720693	-1.288627
19	C	2.798431	-3.576708	-1.257576
20	C	1.562406	-3.112009	-0.812702
21	N	2.416484	0.386903	0.209412
22	C	2.523097	-0.943189	-0.338207
23	C	1.363346	-1.790815	-0.344314
24	B	0.000000	-1.168132	0.000000
25	C	-1.363346	-1.790815	0.344314
26	C	-2.523097	-0.943189	0.338207
27	N	-2.416484	0.386903	-0.209412
28	C	0.000000	3.202138	0.000000
29	C	1.209190	2.509183	0.092723
30	C	1.195228	1.094334	0.117614
31	C	0.000000	0.371032	0.000000
32	C	-1.195228	1.094334	-0.117614
33	C	-1.209190	2.509183	-0.092723
34	Te	-5.391290	-0.063430	0.974232
35	Te	5.391290	-0.063430	-0.974232
36	H	-6.759814	1.284295	-1.542137
37	H	-5.982138	2.366354	-3.629219
38	H	-3.538141	2.444558	-4.145572
39	H	-1.908980	1.478503	-2.556374
40	H	1.908980	1.478503	2.556374
41	H	3.538141	2.444558	4.145572
42	H	5.982138	2.366354	3.629219
43	H	6.759814	1.284295	1.542137
44	H	-0.704656	-3.777405	0.851105
45	H	-2.905137	-4.603879	1.597620
46	H	-4.864617	-3.069533	1.643438
47	H	4.864617	-3.069534	-1.643438
48	H	2.905137	-4.603879	-1.597620
49	H	0.704656	-3.777405	-0.851105
50	H	0.000000	4.287743	0.000000
51	H	2.151111	3.046410	0.133236
52	H	-2.151111	3.046410	-0.133236

Supplementary Table 5 | Nuclear coordinates of S₁ geometry of BN₂PoPo in gas phase optimised using the TD-TPSSH method. For the H, B, C, and N atoms, the 6-31G(d) basis set was used. For the Po atoms, the Stuttgart/Dresden pseudopotentials and basis set (SDD) were used.

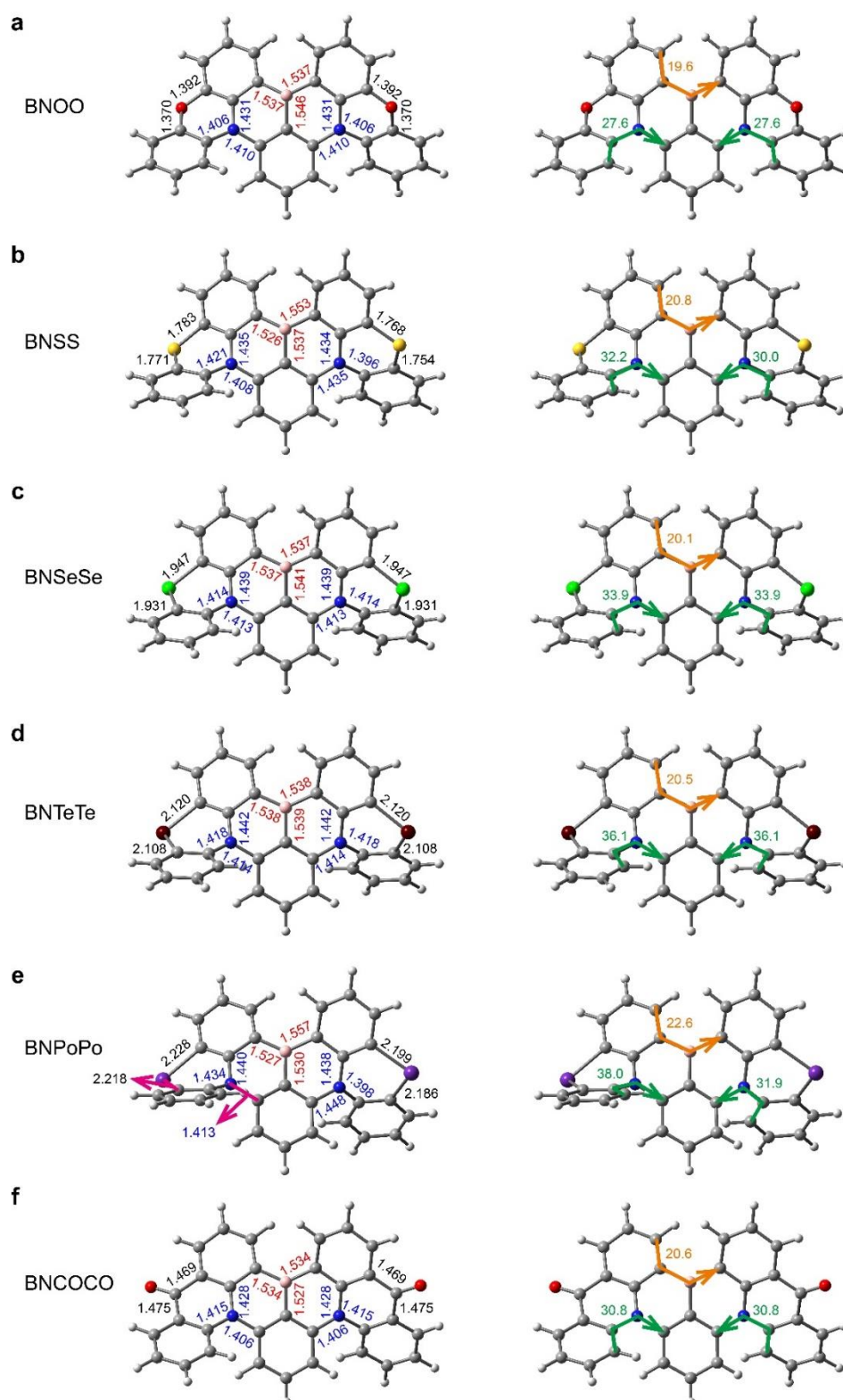
Atom	Element symbol	x (Å)	y (Å)	z (Å)
1	C	-4.782300	-0.592113	1.251140
2	C	-5.663829	-0.827989	2.312850
3	C	-5.174519	-1.073153	3.596607
4	C	-3.794939	-1.037867	3.826902

5	C	-2.916155	-0.771558	2.780102
6	C	2.668356	-1.990067	-2.058160
7	C	3.470808	-2.678496	-2.949948
8	C	4.872380	-2.563401	-2.899216
9	C	5.450151	-1.737767	-1.944454
10	C	4.646995	-1.031230	-1.033607
11	C	-3.395155	-0.567250	1.471304
12	C	3.229186	-1.138976	-1.067597
13	C	-1.552199	2.974020	-1.150174
14	C	-2.789842	3.389491	-1.625511
15	C	-3.903986	2.549203	-1.510356
16	C	-3.753379	1.299247	-0.888765
17	C	3.781130	1.440434	0.552803
18	C	3.953681	2.807135	0.856573
19	C	2.847732	3.645065	0.786687
20	C	1.596937	3.118530	0.441175
21	N	2.375742	-0.469191	-0.186097
22	C	2.542210	0.912239	0.175652
23	C	1.385923	1.761988	0.133888
24	B	-0.012524	1.121627	-0.109957
25	C	-1.360444	1.727105	-0.494789
26	C	-2.532748	0.908820	-0.330911
27	N	-2.457409	-0.321679	0.414146
28	C	-0.048022	-3.122237	0.838552
29	C	1.170404	-2.481872	0.546538
30	C	1.142797	-1.137959	0.172762
31	C	-0.041663	-0.384841	0.155097
32	C	-1.246050	-1.046985	0.458245
33	C	-1.250044	-2.422377	0.779792
34	Po	-5.449926	-0.142606	-0.815616
35	Po	5.574887	0.167615	0.541566
36	H	-6.735869	-0.821664	2.135337
37	H	-5.864756	-1.272077	4.412018
38	H	-3.402199	-1.196340	4.827600
39	H	-1.847237	-0.718853	2.962904
40	H	1.588964	-2.063372	-2.121876
41	H	3.005070	-3.297201	-3.711324
42	H	5.496720	-3.103559	-3.604177
43	H	6.531025	-1.632995	-1.904470
44	H	-0.684293	3.607283	-1.316258
45	H	-2.893164	4.356534	-2.112706
46	H	-4.869257	2.855119	-1.902351
47	H	4.932140	3.198146	1.119794
48	H	2.954744	4.704136	1.005916
49	H	0.738935	3.784119	0.416283
50	H	-0.050632	-4.177267	1.095737
51	H	2.106596	-3.030494	0.590767
52	H	-2.189707	-2.927114	0.981350

Supplementary Table 6 | Nuclear coordinates of S₁ geometry of BNCOCO in gas phase optimised at the TD-TPSSh/6-31G(d) level of theory.

Atom	Element symbol	x (Å)	y (Å)	z (Å)
1	C	-0.048300	4.859033	0.570623
2	C	0.359951	6.031304	1.223221
3	C	1.289809	5.988591	2.257187
4	C	1.855451	4.758720	2.624923
5	C	1.452213	3.580660	2.001920
6	C	-1.452213	-3.580660	2.001920
7	C	-1.855451	-4.758720	2.624923
8	C	-1.289809	-5.988591	2.257187
9	C	-0.359951	-6.031304	1.223221
10	C	0.048300	-4.859033	0.570623
11	C	0.455376	3.612671	1.005296
12	C	-0.455376	-3.612671	1.005296
13	C	-0.936319	1.474792	-3.104385
14	C	-1.423890	2.689509	-3.587524
15	C	-1.428731	3.812528	-2.766092
16	C	-0.934864	3.731489	-1.447809
17	C	0.934864	-3.731489	-1.447809
18	C	1.428731	-3.812528	-2.766092
19	C	1.423890	-2.689509	-3.587524
20	C	0.936319	-1.474792	-3.104385
21	N	0.000000	-2.424326	0.386795
22	C	0.468054	-2.490506	-0.960449
23	C	0.452613	-1.324365	-1.779277
24	B	0.000000	0.000000	-1.151837
25	C	-0.452613	1.324365	-1.779277
26	C	-0.468054	2.490506	-0.960449
27	N	0.000000	2.424326	0.386795
28	C	0.000000	0.000000	3.202583
29	C	0.095612	-1.208684	2.514391
30	C	0.007884	-1.210699	1.097326
31	C	0.000000	0.000000	0.375336
32	C	-0.007884	1.210699	1.097326
33	C	-0.095612	1.208684	2.514391
34	H	-0.063887	6.966108	0.869304
35	H	1.603992	6.903576	2.750507
36	H	2.630820	4.717631	3.384281
37	H	1.931195	2.641670	2.257691
38	H	-1.931195	-2.641670	2.257691
39	H	-2.630820	-4.717631	3.384281
40	H	-1.603992	-6.903576	2.750507
41	H	0.063887	-6.966108	0.869304
42	H	-0.956216	0.598692	-3.745895
43	H	-1.799744	2.758031	-4.604993
44	H	-1.796598	4.776282	-3.101753
45	H	1.796598	-4.776282	-3.101753

46	H	1.799744	-2.758031	-4.604993
47	H	0.956216	-0.598692	-3.745895
48	H	0.000000	0.000000	4.287953
49	H	0.242303	-2.133480	3.059394
50	H	-0.242303	2.133480	3.059394
51	C	0.905240	-4.949405	-0.626994
52	C	-0.905240	4.949405	-0.626994
53	O	1.457783	-6.005610	-0.963549
54	O	-1.457783	6.005610	-0.963549



Supplementary Figure 1 | Ball-and-stick representations of the optimised S_1 geometries of (a) BNOO, (b) BNSS, (c) BNSeSe, (d) BNTeTe, (e) BNPoPo, and (f) BNCOCO. The N-C, B-C, and X-C (X = O, S, Se, Te, Po, and CO) bond lengths are in Å (left). The dihedral angles are in degree (right).

Supplementary Table 7 | S_1 , S_2 , T_1 , T_2 , and T_3 of BNOO calculated using the TD-B3LYP//TD-TPSSh method. Φ_a^r is a Slater determinant that denotes the single-electron excitation from the a th occupied orbital to the r th unoccupied orbital.

State	Excitation energy (eV)	Transition dipole moment (au)	Slater determinant	Contribution (%)
T_1	2.0902		$\Phi_{\text{HOMO}}^{\text{LUMO}}$	97.55
T_2	2.4251		$\Phi_{\text{HOMO}-1}^{\text{LUMO}}$	93.36
			$\Phi_{\text{HOMO}}^{\text{LUMO}+5}$	1.28
S_1	2.5052	1.702	$\Phi_{\text{HOMO}}^{\text{LUMO}}$	98.82
T_3	2.8443		$\Phi_{\text{HOMO}}^{\text{LUMO}+1}$	79.83
			$\Phi_{\text{HOMO}}^{\text{LUMO}+7}$	3.89
			$\Phi_{\text{HOMO}-3}^{\text{LUMO}+7}$	1.98
			$\Phi_{\text{HOMO}-1}^{\text{LUMO}+2}$	1.72
			$\Phi_{\text{HOMO}-2}^{\text{LUMO}+1}$	1.15
			$\Phi_{\text{HOMO}-5}^{\text{LUMO}+1}$	1.06
			$\Phi_{\text{HOMO}-4}^{\text{LUMO}}$	1.06
S_2	2.9575	0.909	$\Phi_{\text{HOMO}-1}^{\text{LUMO}}$	98.58

Supplementary Table 8 | S_1 , S_2 , T_1 , T_2 , and T_3 of BNSS calculated using the TD-B3LYP//TD-TPSSh method. Φ_a^r is a Slater determinant that denotes the single-electron excitation from the a th occupied orbital to the r th unoccupied orbital.

State	Excitation energy (eV)	Transition dipole moment (au)	Slater determinant	Contribution (%)
T_1	2.0879		$\Phi_{\text{HOMO}}^{\text{LUMO}}$	96.97
S_1	2.4388	1.607	$\Phi_{\text{HOMO}}^{\text{LUMO}}$	99.10
T_2	2.4742		$\Phi_{\text{HOMO}-1}^{\text{LUMO}}$	92.93
			$\Phi_{\text{HOMO}}^{\text{LUMO}+5}$	1.46
T_3	2.8686		$\Phi_{\text{HOMO}}^{\text{LUMO}+1}$	51.13
			$\Phi_{\text{HOMO}}^{\text{LUMO}+3}$	10.61
			$\Phi_{\text{HOMO}-2}^{\text{LUMO}}$	6.29
			$\Phi_{\text{HOMO}}^{\text{LUMO}+2}$	5.67
			$\Phi_{\text{HOMO}-1}^{\text{LUMO}+3}$	3.73
			$\Phi_{\text{HOMO}}^{\text{LUMO}+6}$	1.61
			$\Phi_{\text{HOMO}-5}^{\text{LUMO}+1}$	1.23
			$\Phi_{\text{HOMO}-2}^{\text{LUMO}+1}$	1.16
			$\Phi_{\text{HOMO}-1}^{\text{LUMO}+5}$	1.16

			$\Phi_{\text{HOMO}-6}^{\text{LUMO}}$	1.04
S ₂	2.9046	0.892	$\Phi_{\text{HOMO}-1}^{\text{LUMO}}$	98.72

Supplementary Table 9 | S₁, S₂, T₁, T₂, and T₃ of BNSeSe calculated using the TD-B3LYP//TD-TPSSh method. Φ_a^r is a Slater determinant that denotes the single-electron excitation from the *a*th occupied orbital to the *r*th unoccupied orbital.

State	Excitation energy (eV)	Transition dipole moment (au)	Slater determinant	Contribution (%)
T ₁	2.1630		$\Phi_{\text{HOMO}}^{\text{LUMO}}$	97.19
S ₁	2.5105	1.653	$\Phi_{\text{HOMO}}^{\text{LUMO}}$	99.03
T ₂	2.5557		$\Phi_{\text{HOMO}-1}^{\text{LUMO}}$	90.09
			$\Phi_{\text{HOMO}}^{\text{LUMO}+5}$	2.04
			$\Phi_{\text{HOMO}-3}^{\text{LUMO}}$	1.98
			$\Phi_{\text{HOMO}-7}^{\text{LUMO}}$	1.18
T ₃	2.9395		$\Phi_{\text{HOMO}}^{\text{LUMO}+1}$	69.04
			$\Phi_{\text{HOMO}-2}^{\text{LUMO}+1}$	2.87
			$\Phi_{\text{HOMO}}^{\text{LUMO}+5}$	2.74
			$\Phi_{\text{HOMO}-4}^{\text{LUMO}}$	2.60
			$\Phi_{\text{HOMO}}^{\text{LUMO}+7}$	2.52
			$\Phi_{\text{HOMO}-1}^{\text{LUMO}}$	2.44
			$\Phi_{\text{HOMO}-1}^{\text{LUMO}+6}$	1.76
			$\Phi_{\text{HOMO}}^{\text{LUMO}+11}$	1.72
			$\Phi_{\text{HOMO}-3}^{\text{LUMO}+6}$	1.47
			$\Phi_{\text{HOMO}-8}^{\text{LUMO}+11}$	1.46
			$\Phi_{\text{HOMO}-5}^{\text{LUMO}+1}$	1.11
S ₂	2.9543	0.498	$\Phi_{\text{HOMO}-1}^{\text{LUMO}}$	98.92

Supplementary Table 10 | S₁, S₂, T₁, T₂, and T₃ of BNTeTe calculated using the TD-B3LYP//TD-TPSSh method. Φ_a^r is a Slater determinant that denotes the single-electron excitation from the *a*th occupied orbital to the *r*th unoccupied orbital.

State	Excitation energy (eV)	Transition dipole moment (au)	Slater determinant	Contribution (%)
T ₁	2.1827		$\Phi_{\text{HOMO}}^{\text{LUMO}}$	96.36
			$\Phi_{\text{HOMO}-2}^{\text{LUMO}}$	1.34
S ₁	2.5158	1.6154	$\Phi_{\text{HOMO}}^{\text{LUMO}}$	99.01

T ₂	2.5744		$\Phi_{\text{HOMO}-1}^{\text{LUMO}}$	88.03
			$\Phi_{\text{HOMO}-3}^{\text{LUMO}}$	4.46
			$\Phi_{\text{HOMO}}^{\text{LUMO}+5}$	1.84
			$\Phi_{\text{HOMO}-7}^{\text{LUMO}}$	1.21
T ₃	2.8780		$\Phi_{\text{HOMO}}^{\text{LUMO}+2}$	26.51
			$\Phi_{\text{HOMO}-1}^{\text{LUMO}+1}$	24.82
			$\Phi_{\text{HOMO}-2}^{\text{LUMO}}$	23.49
			$\Phi_{\text{HOMO}-2}^{\text{LUMO}+2}$	8.27
			$\Phi_{\text{HOMO}-5}^{\text{LUMO}}$	4.27
			$\Phi_{\text{HOMO}-6}^{\text{LUMO}}$	1.65
S ₂	2.9086	0.6065	$\Phi_{\text{HOMO}-1}^{\text{LUMO}}$	98.44

Supplementary Table 11 | S₁, S₂, T₁, T₂, and T₃ of BNPoPo calculated using the TD-B3LYP//TD-TPSSh method. Φ_a^r is a Slater determinant that denotes the single-electron excitation from the *a*th occupied orbital to the *r*th unoccupied orbital.

State	Excitation energy (eV)	Transition dipole moment (au)	Slater determinant	Contribution (%)
T ₁	2.1339		$\Phi_{\text{HOMO}}^{\text{LUMO}}$	92.84
			$\Phi_{\text{HOMO}-2}^{\text{LUMO}}$	2.31
			$\Phi_{\text{HOMO}-1}^{\text{LUMO}}$	2.15
S ₁	2.4062	1.3456	$\Phi_{\text{HOMO}}^{\text{LUMO}}$	99.23
T ₂	2.5265		$\Phi_{\text{HOMO}-1}^{\text{LUMO}}$	84.24
			$\Phi_{\text{HOMO}-3}^{\text{LUMO}}$	5.28
			$\Phi_{\text{HOMO}-2}^{\text{LUMO}+2}$	2.90
			$\Phi_{\text{HOMO}}^{\text{LUMO}}$	2.15
T ₃	2.6655		$\Phi_{\text{HOMO}}^{\text{LUMO}+2}$	81.27
			$\Phi_{\text{HOMO}-1}^{\text{LUMO}+2}$	9.07
			$\Phi_{\text{HOMO}-2}^{\text{LUMO}+2}$	4.75
S ₂	2.8286	0.9483	$\Phi_{\text{HOMO}-1}^{\text{LUMO}}$	98.58

Supplementary Table 12 | S₁, S₂, T₁, T₂, and T₃ of BNCOCO calculated using the TD-B3LYP//TD-TPSSh method. Φ_a^r is a Slater determinant that denotes the single-electron excitation from the *a*th occupied orbital to the *r*th unoccupied orbital.

State	Excitation energy (eV)	Transition dipole moment (au)	Slater determinant	Contribution (%)
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T ₁	2.2005		$\Phi_{\text{HOMO}}^{\text{LUMO}}$	95.56
			$\Phi_{\text{HOMO}-1}^{\text{LUMO}+1}$	2.22
T ₂	2.5191		$\Phi_{\text{HOMO}}^{\text{LUMO}+1}$	51.81
			$\Phi_{\text{HOMO}-1}^{\text{LUMO}}$	33.11
			$\Phi_{\text{HOMO}}^{\text{LUMO}+3}$	4.78
			$\Phi_{\text{HOMO}-1}^{\text{LUMO}+2}$	1.58
			$\Phi_{\text{HOMO}-5}^{\text{LUMO}+1}$	1.25
S ₁	2.5454	2.2007	$\Phi_{\text{HOMO}}^{\text{LUMO}}$	98.61
T ₃	2.8368		$\Phi_{\text{HOMO}}^{\text{LUMO}+2}$	33.91
			$\Phi_{\text{HOMO}-1}^{\text{LUMO}+1}$	26.84
			$\Phi_{\text{HOMO}-5}^{\text{LUMO}}$	16.86
			$\Phi_{\text{HOMO}-3}^{\text{LUMO}}$	4.17
			$\Phi_{\text{HOMO}-11}^{\text{LUMO}+1}$	1.83
			$\Phi_{\text{HOMO}-10}^{\text{LUMO}}$	1.57
			$\Phi_{\text{HOMO}-4}^{\text{LUMO}+1}$	1.11
			$\Phi_{\text{HOMO}-6}^{\text{LUMO}}$	1.10
			$\Phi_{\text{HOMO}-2}^{\text{LUMO}+1}$	1.08
			$\Phi_{\text{HOMO}-8}^{\text{LUMO}}$	1.08
S ₂	3.1061	0.2341	$\Phi_{\text{HOMO}}^{\text{LUMO}+1}$	91.50
			$\Phi_{\text{HOMO}-1}^{\text{LUMO}}$	5.47

Supplementary Method 4

T₁-energy correction with the combined TD-B3LYP and TDA-B2-PLYP method

First, we calculated the S₁ and T₂ energies (denoted $E_{\text{B3LYP}}(\text{S}_1)$ and $E_{\text{B3LYP}}(\text{T}_2)$) using the TD-B3LYP/6-31G(d) (and SDD) method for the optimised S₁ geometry. Then, we calculated $\Delta E(\text{T}_1 \rightarrow \text{S}_1)$ (denoted as $\Delta E_{\text{B2-PLYP}}(\text{T}_1 \rightarrow \text{S}_1)$) using the TDA-B2-PLYP/def2-TZVP method for the same geometry. Finally, we calculated the T₁ energy as $E(\text{T}_1) = E_{\text{B3LYP}}(\text{S}_1) - \Delta E_{\text{B2-PLYP}}(\text{T}_1 \rightarrow \text{S}_1)$ (Table S13). For the S₁ and T₂ energies, we used the TD-B3LYP/6-31G(d) results without correction: $E(\text{S}_1) = E_{\text{B3LYP}}(\text{S}_1)$ and $E(\text{T}_2) = E_{\text{B3LYP}}(\text{T}_2)$. We used the $E(\text{S}_1)$, $E(\text{T}_1)$, and $E(\text{T}_2)$ values to calculate the rate constants.

Supplementary Table 13 | Calculated excited-state energies and energy differences with the combined TD-B3LYP and TDA-B2-PLYP method in eV. The TD-B3LYP calculations were performed with Gaussian 16 Rev C01 program package.³ The TDA-B2-PLYP calculations were performed with ORCA 5.0.3 program package.⁶⁻⁸

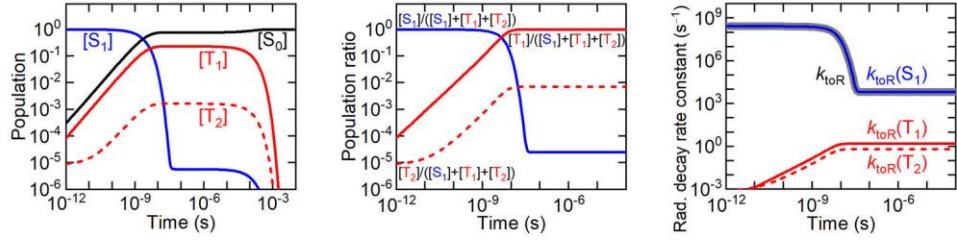
	BNOO	BNSS	BNSeSe	BNTeTe	BNPoPo	BNCOCO
$E(S_1) = E_{B3LYP}(S_1)$	2.51	2.44	2.51	2.52	2.41	2.55
$E(T_2) = E_{B3LYP}(T_2)$	2.43	2.47	2.56	2.57	2.53	2.52
$\Delta E_{B3LYP}(T_1 \rightarrow S_1)$	0.42	0.35	0.35	0.33	0.27	0.35
$\Delta E(T_1 \rightarrow S_1) = \Delta E_{B2-PLYP}(T_1 \rightarrow S_1)$	0.21	0.14	0.14	0.14	0.14	0.14
$E(T_1) = E(S_1) - \Delta E(T_1 \rightarrow S_1)$	2.30	2.30	2.37	2.38	2.26	2.41
$\Delta E(T_2 \rightarrow S_1) = E(S_1) - E(T_2)$	0.08	-0.04	-0.05	-0.06	-0.12	0.03
$\Delta E(T_1 \rightarrow T_2) = E(T_2) - E(T_1)$	0.13	0.18	0.19	0.20	0.26	0.11
Experiments						
$E(S_1)$	2.54*	2.55*	2.58*			
$E(T_1)$	2.39*	2.42*	2.44*			
$\Delta E(T_1 \rightarrow S_1)$	0.15*	0.13*	0.14*			

*) Experimental data from ref. 9.

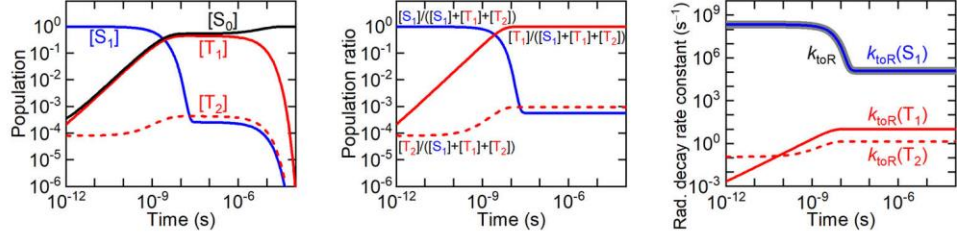
Supplementary Table 14 | Calculated excited-state energies and energy differences with the ADC(2) and SCS-CC2 methods in eV. The def2-TZVP basis set was used for all the atoms. The ADC(2) and SCS-CC2 calculations were performed with TURBOMOLE v7.4.1 2019 program package.¹⁰

	BNOO	BNSS	BNSeSe	BNTeTe	BNPoPo	BNCOCO
ADC(2)						
$E_{\text{ADC2}}(\text{S}_1)$	2.84	2.30	2.35	2.35	2.33	3.07
$E_{\text{ADC2}}(\text{T}_1)$	2.61	2.22	2.27	2.27	2.25	2.83
$E_{\text{ADC2}}(\text{T}_2)$	3.14	2.67	3.23	2.77	2.71	3.03
$\Delta E_{\text{ADC2}}(\text{T}_1 \rightarrow \text{S}_1)$	0.23	0.09	0.08	0.08	0.07	0.24
$\Delta E_{\text{ADC2}}(\text{T}_2 \rightarrow \text{S}_1)$	-0.30	-0.37	-0.88	-0.42	-0.39	0.04
$\Delta E_{\text{ADC2}}(\text{T}_1 \rightarrow \text{T}_2)$	0.53	0.45	0.96	0.49	0.46	0.20
SCS-CC2						
$E_{\text{SCS-CC2}}(\text{S}_1)$	2.68	2.68	2.72	2.72	2.72	3.36
$E_{\text{SCS-CC2}}(\text{T}_1)$	2.58	2.58	2.63	2.64	2.64	3.09
$E_{\text{SCS-CC2}}(\text{T}_2)$	3.02	3.02	3.11	3.12	2.97	3.49
$\Delta E_{\text{SCS-CC2}}(\text{T}_1 \rightarrow \text{S}_1)$	0.10	0.10	0.09	0.09	0.08	0.27
$\Delta E_{\text{SCS-CC2}}(\text{T}_2 \rightarrow \text{S}_1)$	-0.34	-0.34	-0.39	-0.39	-0.25	-0.13
$\Delta E_{\text{SCS-CC2}}(\text{T}_1 \rightarrow \text{T}_2)$	0.43	0.43	0.48	0.48	0.33	0.40

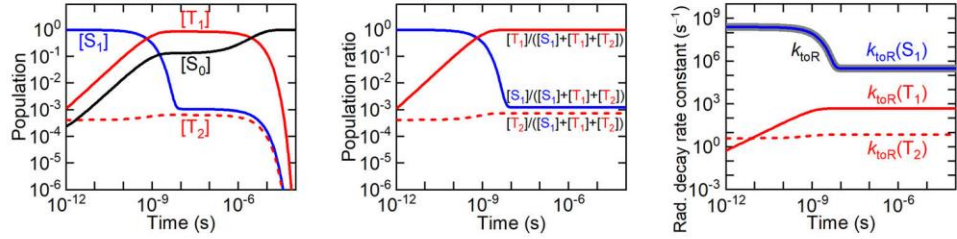
BNOO



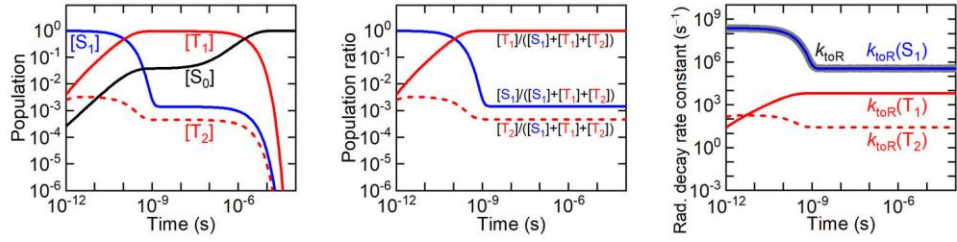
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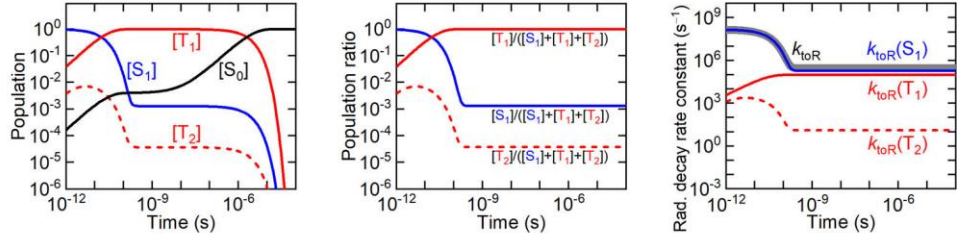
BNSeSe



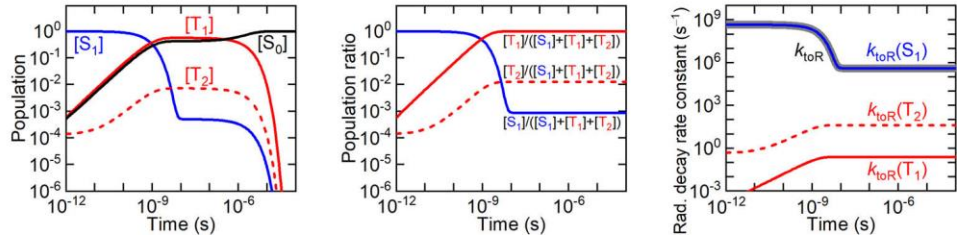
BNTeTe



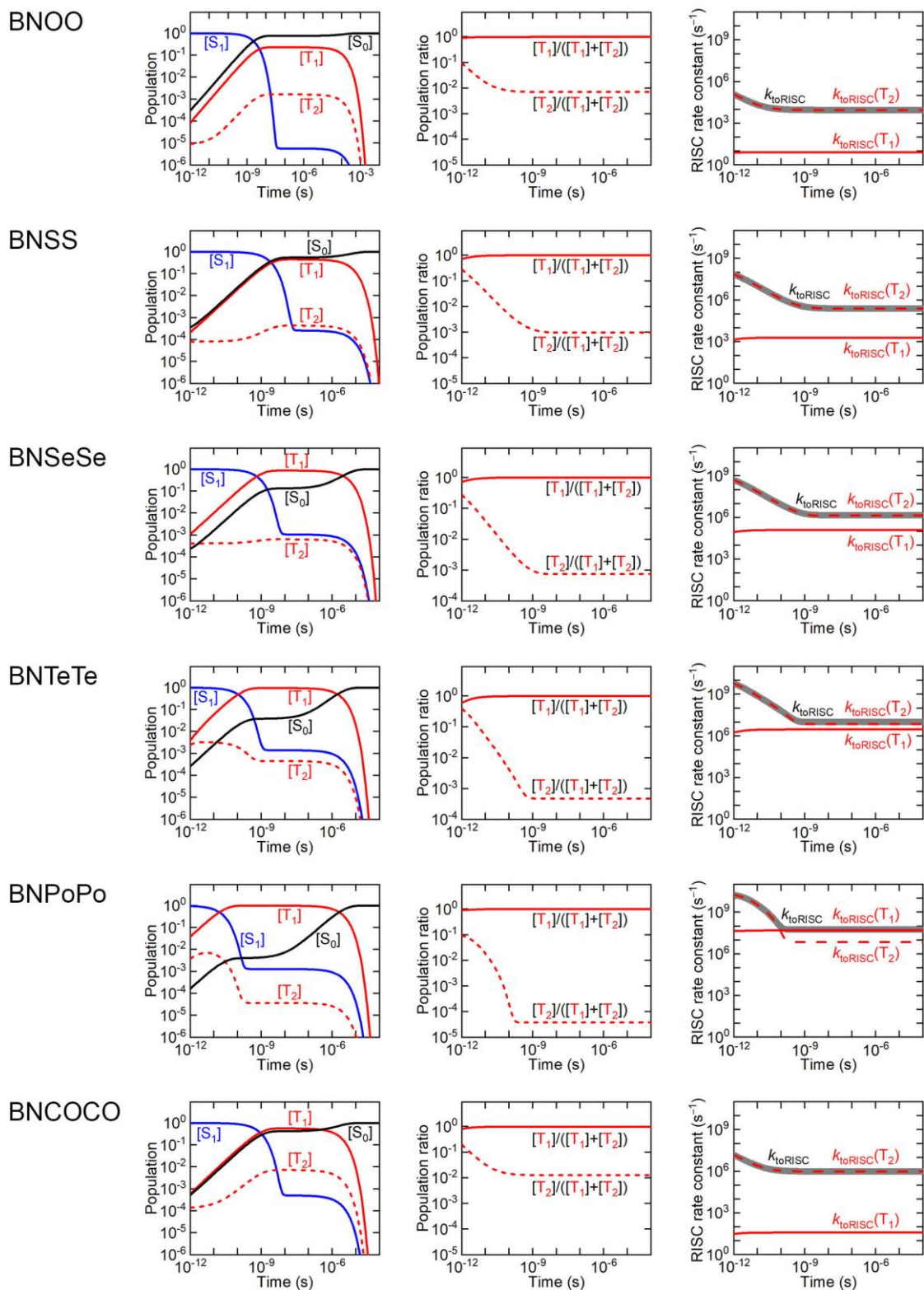
BNPoPo



BNCOCO



Supplementary Figure 2 | Calculated population, population ratio, and rate constants for total radiative decay for BNOO, BNSS, BNSeSe, BNTeTe, BNPoPo, and BNCOCO. $k_{toR}(S_1)$, $k_{toR}(T_1)$, and $k_{toR}(T_2)$ are the contributions from $S_1 \rightarrow S_0$ fluorescence including TADF, $T_1 \rightarrow S_0$ phosphorescence, and $T_2 \rightarrow S_0$ phosphorescence to k_{toR} , respectively: $k_{toR} = k_{toR}(S_1) + k_{toR}(T_1) + k_{toR}(T_2)$; $k_{toR}(S_1) = k_F(S_1 \rightarrow S_0) \times [S_1]/([S_1] + [T_1] + [T_2])$; $k_{toR}(T_1) = k_{Phos}(T_1 \rightarrow S_0) \times [T_1]/([S_1] + [T_1] + [T_2])$; $k_{toR}(T_2) = k_{Phos}(T_2 \rightarrow S_0) \times [T_2]/([S_1] + [T_1] + [T_2])$. k_{toR} for BNOO/BNSS/BNSeSe/BNTeTe/BNPoPo/ BNCOCO is constant in the time domain longer than 100/100/10/10/1/10 ns.



Supplementary Figure 3 | Calculated population, population ratio, and rate constants for reverse intersystem crossing (RISC) for BNOO, BNSS, BNSeSe, BNTeTe, BNPoPo, and BNCOCO. $k_{\text{toRISC}}(T_1)$ and $k_{\text{toRISC}}(T_2)$ are the contributions from $T_1 \rightarrow S_1$ and $T_2 \rightarrow S_1$ RISCs to k_{toRISC} , respectively: $k_{\text{toRISC}} = k_{\text{toRISC}}(T_1) + k_{\text{toRISC}}(T_2)$; $k_{\text{toRISC}}(T_1) = k_{\text{RISC}}(T_1 \rightarrow S_1) \times [T_1] / ([T_1] + [T_2])$; $k_{\text{toRISC}}(T_2) = k_{\text{RISC}}(T_2 \rightarrow S_1) \times [T_2] / ([T_1] + [T_2])$. k_{toRISC} for BNOO/BNSS/BNSeSe/BNTeTe/BNPoPo/BNCOCO is constant in the time domain longer than 1/10/10/10/1/10 ns.

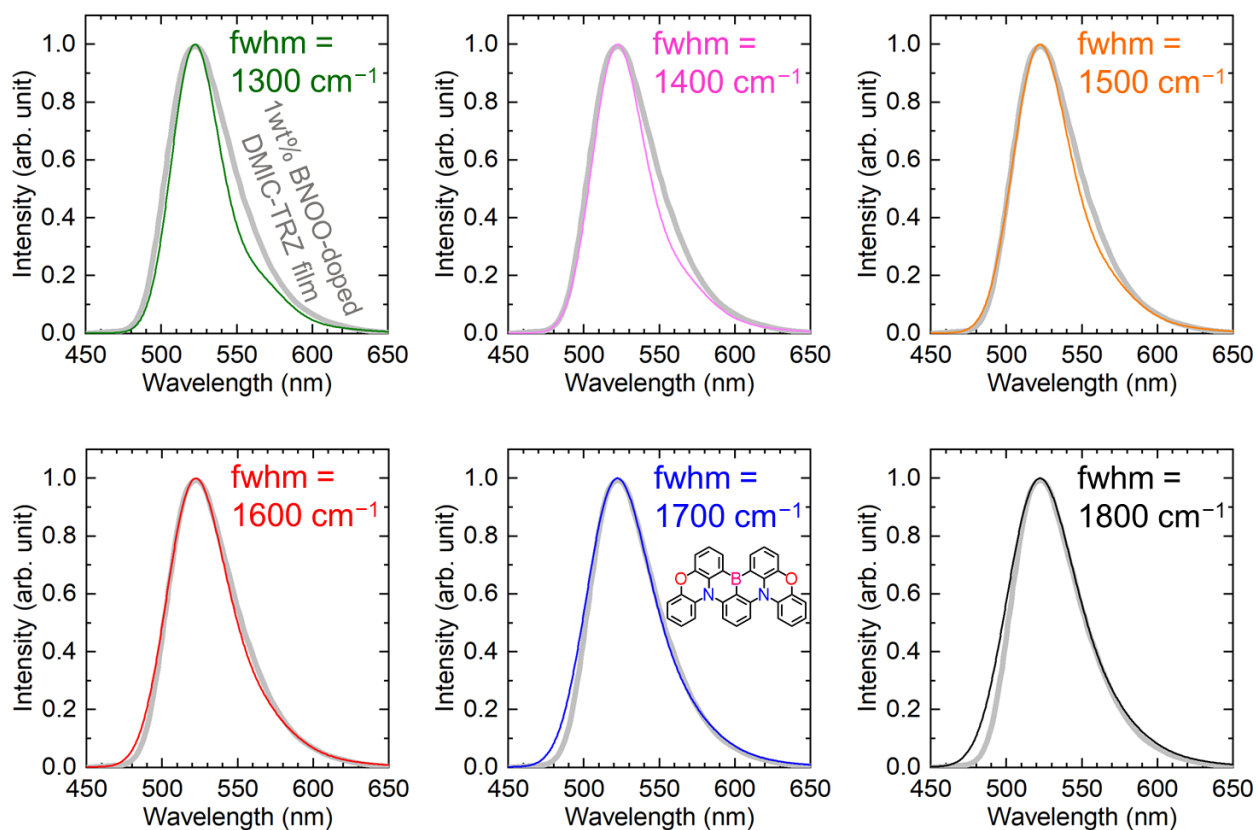
Supplementary Method 5

Calculation of emission spectra

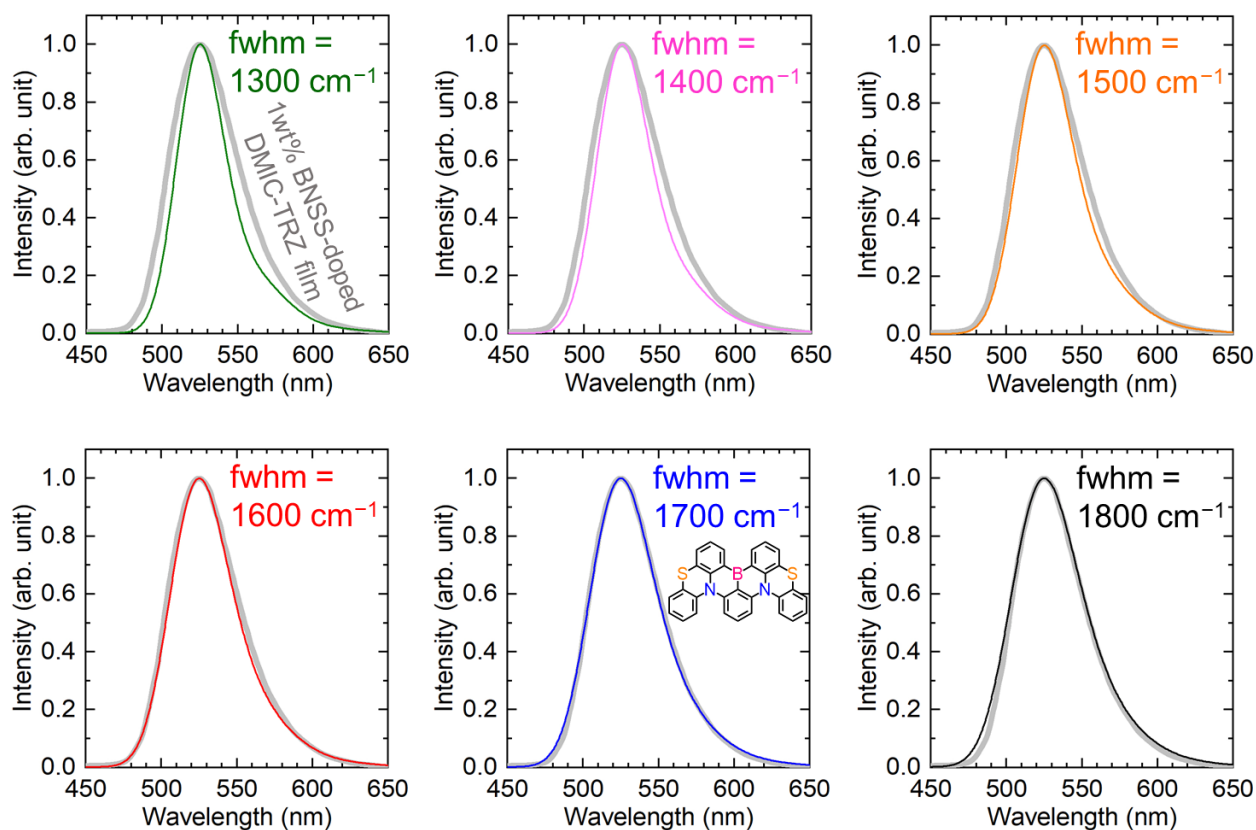
The emission spectra were calculated using the vertical gradient method implemented in ORCA 5.0.3 program package⁶⁻⁸ for the S_0 geometry optimised at the PBE0/Def2-SVP level. Frequency analysis was also performed at the same level. The input file for the geometry optimisation, frequency analysis, and spectrum calculations were as follows.

```
# Geometry optimisation and frequency analysis
!PBE0 Def2-SVP Opt Freq
* XYZFILE 0 1 S0_init.xyz

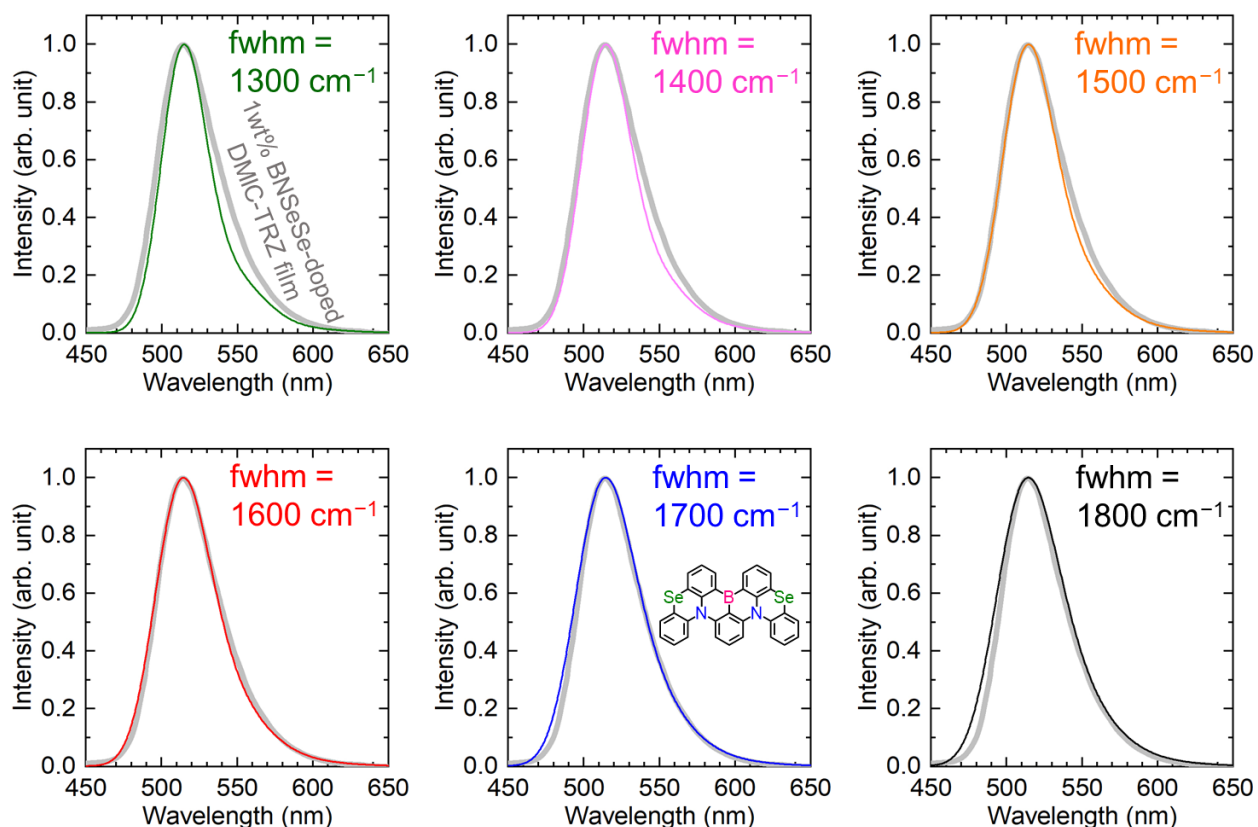
# Spectrum simulation
!PBE0 DEF2-SVP TIGHTSCF ESD(FLUOR)
%TDDFT NROOTS 1
IROOT 1
END
%ESD
GSHESSIAN "S0.hess" # Use the Hessian calculated at the optimised S0-state geometry
DOHT TRUE
HESSFLAG VG # Use the vertical gradient method
LINES Gauss # Use the Gaussian distribution function
INLINEW 679.40552
SPECRES 5.0 # Spectral resolution
UNIT NM
END
* XYZFILE 0 1 S0.xyz # Use the optimised S0-state geometry
```



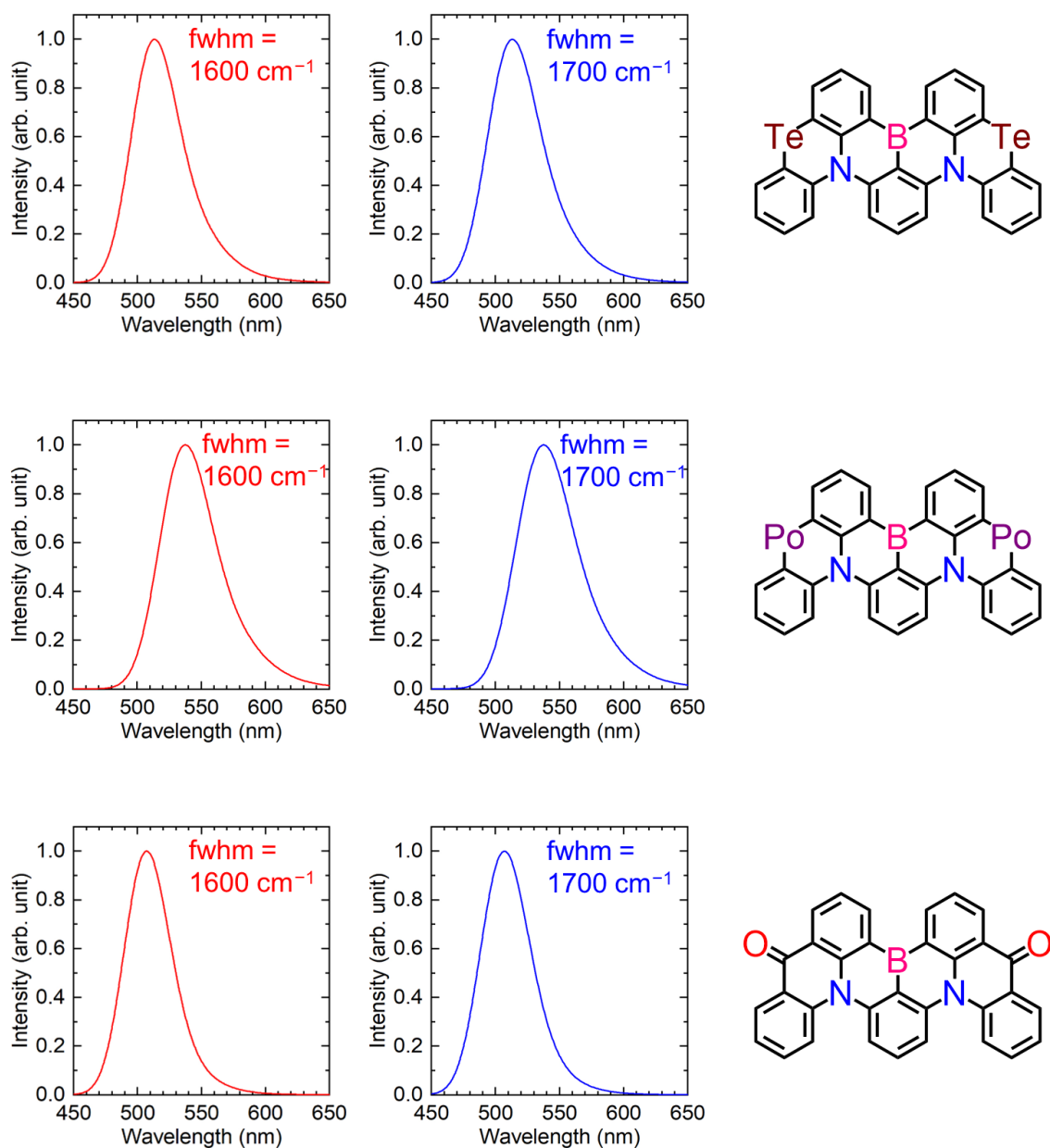
Supplementary Figure 4 | Experimental and calculated emission spectra for BNOO. The grey curve shows the experimental spectrum for 1 wt% BNOO-doped DMIC-TRZ film; the coloured curves show the calculated emission spectra. The Gaussian distribution functions with FWHM values of 1300 (green), 1400 (pink), 1500 (orange), 1600 (red), 1700 (blue), and 1800 cm^{-1} (black) were examined to reproduce the experimental spectrum. The best fit was obtained for the FWHM values of 1600 cm^{-1} (red).



Supplementary Figure 5 | Experimental and calculated emission spectra for BNSS. The grey curve shows the experimental spectrum for 1 wt% BNSS-doped DMIC-TRZ film; the coloured curves show the calculated emission spectra. The Gaussian distribution functions with FWHM values of 1300 (green), 1400 (pink), 1500 (orange), 1600 (red), 1700 (blue), and 1800 cm^{-1} (black) were examined to reproduce the experimental spectrum. The best fit was obtained for the FWHM values of 1700 cm^{-1} (red).



Supplementary Figure 6 | Experimental and calculated emission spectra for BNSeSe. The grey curve shows the experimental spectrum for 1 wt% BNSeSe-doped DMIC-TRZ film; the coloured curves show the calculated emission spectra. The Gaussian distribution functions with FWHM values of 1300 (green), 1400 (pink), 1500 (orange), 1600 (red), 1700 (blue), and 1800 cm⁻¹ (black) were examined to reproduce the experimental spectrum. The best fit was obtained for the FWHM values of 1600 cm⁻¹ (red).



Supplementary Figure 7 | Calculated emission spectra for BNTeTe, BNPoPo, and BNCOCO. The Gaussian distribution functions with FWHM values of 1600 (red) and 1700 cm^{-1} (blue) were applied.

Supplementary Method 6

Molecular design for minimizing T₁-T₂ energy gap

Supplementary Figure 8a shows a case when T₁ and T₂ consist of HOMO → LUMO and HOMO-1 → LUMO transitions, respectively. The T₁ and T₂ energies can be expressed as¹¹

$$E(T_1) = 2h_{H-1H-1} + h_{HH} + h_{LL} + J_{H-1H-1} + 2J_{H-1H} + 2J_{H-1L} + J_{HL} - K_{H-1H} - K_{HL} \quad (D1)$$

$$E(T_2) = h_{H-1H-1} + 2h_{HH} + h_{LL} + J_{HH} + 2J_{H-1H} + J_{H-1L} + 2J_{HL} - K_{H-1H} - K_{HL} \quad (D2)$$

where h denotes the core integral (kinetic and potential energies), J denotes the Coulomb integral, and K denotes the exchange integral. Then, $\Delta E(T_2 - T_1)$ is written as

$$\Delta E(T_2 - T_1) = (h_{HH} - h_{H-1H-1}) + (J_{HH} - J_{H-1H-1}) + (J_{HL} - J_{H-1L}) \quad (D3)$$

A simple approach of minimizing $\Delta E(T_1 \rightarrow T_2)$ is to decrease the first term $h_{HH} - h_{H-1H-1}$, which is possible by expanding the HOMO and HOMO-1 distributions (the second and third terms of $\Delta E(T_1 \rightarrow T_2)$, expressed in terms of the Coulomb integrals, are not easy to control).

Supplementary Figure 8b shows a case where T₁ and T₂ consist of the HOMO → LUMO and HOMO → LUMO+1 transitions, respectively. The T₁ and T₂ energies can be expressed as

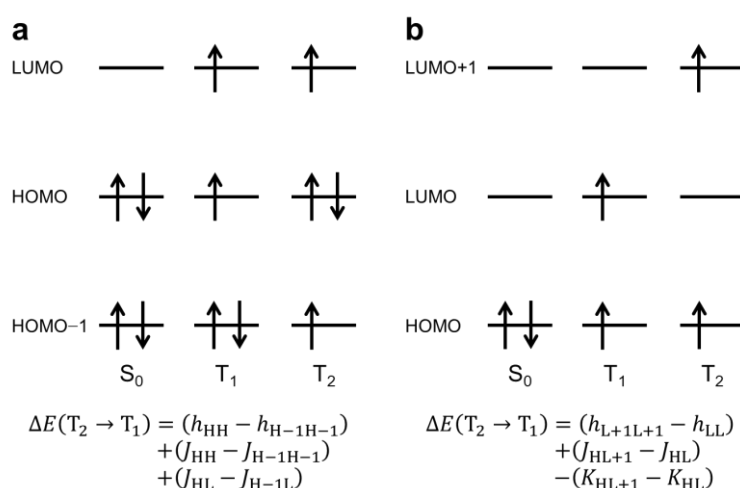
$$E(T_1) = h_{HH} + h_{LL} + J_{HL} - K_{HL} \quad (D4)$$

$$E(T_2) = h_{HH} + h_{L+1L+1} + J_{HL+1} - K_{HL+1} \quad (D5)$$

Hence, $\Delta E(T_2 - T_1)$ is written as

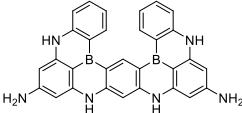
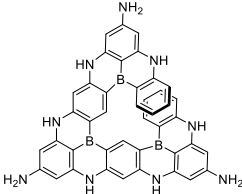
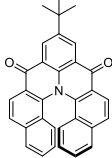
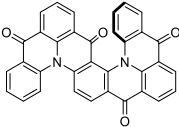
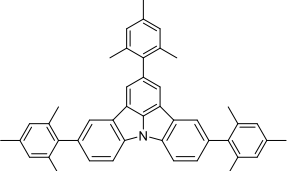
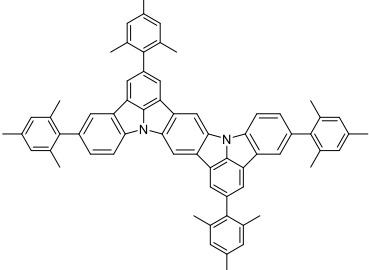
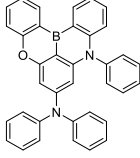
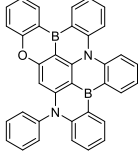
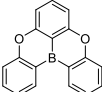
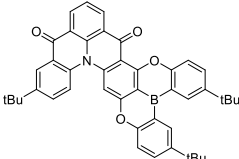
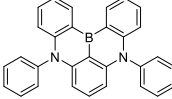
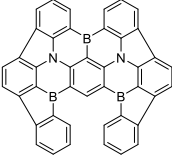
$$\Delta E(T_2 - T_1) = (h_{L+1L+1} - h_{LL}) + (J_{HL+1} - J_{HL}) - (K_{HL+1} - K_{HL}) \quad (D6)$$

For the same reason, expanding the LUMO+1 and LUMO distributions results in an effective approach to minimize $\Delta E(T_1 \rightarrow T_2)$ by decreasing $h_{L+1L+1} - h_{LL}$. Regardless of whether T₂ is described as the HOMO-1 → LUMO or HOMO → LUMO+1 transition, expanding molecular orbitals relevant for T₁ and T₂ is a simple way to decrease $\Delta E(T_1 \rightarrow T_2)$ and accelerate the T₂-mediated RISC process.



Supplementary Figure 8 | S₀, T₁, and T₂ determinants and $\Delta E(T_1 \rightarrow T_2)$. (a) A case when T₁ and T₂ consist of HOMO → LUMO (H → L) and HOMO-1 → LUMO (H-1 → L) transitions, respectively. (b) A case when T₁ and T₂ consist of HOMO → LUMO (H → L) and HOMO → LUMO+1 (H → L+1) transitions, respectively.

Supplementary Table 15 | Minimization of $\Delta E(T_1 \rightarrow T_2)$. These examples demonstrate that expanding molecular orbitals relevant for T_1 and T_2 is an effective way to decrease $\Delta E(T_1 \rightarrow T_2)$, resulting in acceleration of the T_2 -mediated RISC process.

MR-TADF molecules with $\Delta E(T_2 - T_1)$		References (in the main text)
 0.147 eV	 0.098 eV	12 (45)
 0.28 eV	 0.12 eV	13 (46)
 0.46 eV	 0.33 eV	14 (47)
 0.21 eV	 0.07 eV	15 (48)
 0.58 eV	 0.28 eV	16 (49)
 0.81 eV	 0.04 eV	17 (50)

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